

Issues Identified

Capstone 2 — Debug Task Manager

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Review of the starter repository (`app.js`, `dom.js`, `utils.js`, `app.test.js`, `index.html`) before any fixes. Line numbers refer to the original starter files.

Variables, Data Types & Operators

1. `app.js:4` — `taskList` assigned without `let/const` (implicit global).
2. `var` used throughout `app.js` (5, 37, 46, 55, 70, 82–85, 98, 122, 124, 133, 135), `dom.js` and `utils.js`; no `let/const` anywhere.
3. Loose `==` comparisons: `app.js:57`; `utils.js:32`.
4. `app.js:71` — assignment (`=`) inside an `if` condition; always truthy, so the first task always “matches” and its `id` is overwritten.
5. No `typeof` checks; `updateTaskPriority` (`app.js:66`) accepts any input unvalidated.
6. `utils.js:31–36` — `isHighPriority` returns the strings “yes”/“no” instead of booleans.
7. `utils.js:20` — `generateRandomId` returns a raw decimal; collision-prone.
8. `utils.js:24–28` — `formatTaskName` returns its input unchanged (no trim or capitalisation).

Control Flow

9. `app.js:46` — off-by-one (`i <= taskList.length`) reads past the array (`TypeError`).
10. `app.js:55–61` — `while` loop never increments `i` (infinite loop).
11. No `for-of` loops; imperative loops where `filter/find/map` fit (`app.js:96–106`, 124–128, 135–139).
12. `app.js:121–129` — no empty-array check; divides by zero (`NaN`); no rounding.

Functions & Functional Programming

13. `app.js:52` — `findTaskByTitle` omits its `title` parameter (`ReferenceError` when called).
14. `app.js:109–117` — recursion has no base case or null guard (stack overflow past the array end).
15. No pure functions, higher-order functions, or array methods (`map`, `filter`, `reduce`, `find`, `some`, `every`) anywhere.

OOP & Classes

16. `app.js:14` — `Task` has no `id` property.
17. `app.js:17` — `Task` has no `toggleCompletion` method.
18. `app.js:26–31` — `SubTask` omits `super()` (`ReferenceError` on `this`) and never forwards `title`, `description`, `priority`.
19. `app.js:144–153` — `TaskManager` exposes only `getTotalTasks`.

Modern JavaScript

20. String concatenation instead of template literals: `app.js:21`; `dom.js:44–45`.
21. No destructuring (`app.js:82–85`), no spread (`app.js:96–106`), no rest parameters.
22. No ES6 modules — zero `export/import` statements in any file.

DOM & Events

23. `dom.js:6` — `getElementById` given a class selector; `dom.js:7` — `querySelector` missing `#`, and no such element exists in `index.html`.

- 24. `dom.js:10` — listener attached with no null check (crashes on load).
- 25. `dom.js:16–32` — `handleAddTask`: no validation, no `preventDefault`, priority hard-coded (no input in the HTML), fields never cleared.
- 26. `dom.js:35–50` — container never cleared (duplicates on re-render); `innerHTML` written twice; rendered tasks lack ids, status, and `complete/delete` handlers.
- 27. `dom.js:53–61` — no event delegation or `event.target` checks.
- 28. `dom.js:67` — initialisation runs before the DOM is ready (no `DOMContentLoaded`); `index.html` loads `dom.js` before `app.js`.

JSON & Storage

- 29. `utils.js:7–16` — `saveToStorage/loadFromStorage` skip `JSON.stringify/JSON.parse`, so objects persist as “[object Object]”.

Testing

- 30. `app.test.js` — no imports, no `beforeEach` reset, only two shallow tests with a weak assertion; no coverage of inheritance, recursion, error handling, or edge cases.

Error Handling & Code Quality

- 31. Zero `try-catch` blocks and no parameter validation in any file.
- 32. Sparse comments, mixed conventions, redundant duplicated loops.